

CHAPTER 7

7.1) DENSITY OF ORBITALS IN ONE AND TWO DIMENSIONS

(a) In one dimension, $\psi(x) = A \sin(n\pi x/L)$, analogously to (3.58). Here n is an integer. The number of electrons that can be accommodated in the orbitals up to a given quantum number n is $N = 2n$, where the factor 2 arises from the electron spin. The corresponding maximum energy is

$$\varepsilon = \frac{\hbar^2}{2m} \left(\frac{n\pi}{L} \right)^2 = \frac{1}{2m} \left(\frac{\pi \hbar N}{2L} \right)^2 ; N = (2L/\pi \hbar) (2m\varepsilon)^{1/2} . \quad (S1)$$

If N is the total number of electrons actually present, then ε is the Fermi energy ε_F . From (S1):

$$\frac{d\varepsilon}{\varepsilon} = \frac{2dN}{N} , \quad \mathcal{D}_1(\varepsilon) = \frac{dN}{d\varepsilon} = \frac{N}{2\varepsilon} = (L/\pi \hbar) (2m/\varepsilon)^{1/2} .$$

(b) In two dimensions, the relation between the number of electrons N and the highest orbital n is

$$N = 2 \times (\pi n^2/4) , \quad n = (2N/\pi)^{1/2} .$$

Therefore

$$\varepsilon = \frac{\hbar^2}{2m} \left(\frac{n\pi}{L} \right)^2 = \frac{\pi \hbar^2 N}{mL^2} , \quad N = L^2 m \varepsilon / \pi \hbar^2 ,$$

$$\mathcal{D}_2(\varepsilon) = \frac{dN}{d\varepsilon} = L^2 m / \pi \hbar^2 = A m / \pi \hbar^2 .$$

7.2) ENERGY OF RELATIVISTIC FERMI GAS

(a) With the help of (6) -- $n_F = (3N/\pi)^{1/3}$, which remains valid -- we obtain, with $n = N/L^3$:

$$\varepsilon_F = p_F c = (\hbar \pi c / L) n_F = \hbar \pi c (3n/\pi)^{1/3} .$$

(b) With (9):

$$\begin{aligned}
 U(0) &= 2 \sum \varepsilon(n) = \pi \int_0^{n_F} n^2 \varepsilon(n) dn = \pi (\hbar^3 c / L) \int_0^{n_F} n^3 dn \\
 &= \pi (\hbar^3 c / L) n_F^4 / 4 = \pi \varepsilon_F n_F^3 / 4 = (3N/4) \varepsilon_F .
 \end{aligned}$$

7.3) PRESSURE AND ENTROPY OF DEGENERATE FERMI GAS

(a) From (10) and (7):

$$U_0 = \frac{3}{5} N \varepsilon_F, \text{ where } \varepsilon_F = \frac{\hbar^2}{2m} \left(\frac{3\pi^2 N}{V} \right)^{2/3} . \quad (S1)$$

The pressure is obtained from (3.26):

$$p = -(\partial U / \partial V)_{\sigma, N} = -(\partial U_0 / \partial V)_N .$$

In the second equality we have set $U = U_0$ and have dropped the subscript σ , because a volume change on the system while the system is kept in its ground state is a change at constant entropy. Because, from (S1), $U_0 \propto V^{-2/3}$, we have

$$p = \frac{2}{3} \frac{U_0}{V} = \frac{2}{5} \frac{N}{V} \varepsilon_F .$$

Insertion of the expression for ε_F yields (90), but the form given here is more "physical": The pressure is a fixed multiple of the product of particle concentration and Fermi energy.

(b) The fastest way to obtain σ is with the help of the relation (3.17a),

$$C_V = \tau (\partial \sigma / \partial \tau)_V , \quad (S2)$$

from the low-temperature heat capacity of the Fermi gas, already calculated in (37). According to (37), C_V is simply proportional to τ . Because of (S2), together with

the requirement that $\sigma \rightarrow 0$ as $\tau \rightarrow 0$, this implies that σ is equal to C_V :

$$\sigma = C_V = \pi^2 N \tau / 2 \epsilon_F \quad . \quad (S3)$$

An example of an alternate way to obtain σ is the following. From $\sigma = -\partial F / \partial \tau$ and $\mu = \partial F / \partial N$ follow the Maxwell relation

$$\left(\frac{\partial \sigma}{\partial N} \right)_{\tau} = - \left(\frac{\partial \mu}{\partial \tau} \right)_N \quad . \quad (S4)$$

The quantity μ is calculated in the next problem, see (7.4.S4). If (7.4.S4) is inserted into (S3):

$$\left(\frac{\partial \sigma}{\partial N} \right)_{\tau} = \pi^2 \tau / 6 \epsilon_F \quad . \quad (S5)$$

From (7) follows $1/\epsilon_F \propto N^{-2/3}$. Integrating this is the same as multiplying by $3N$,

$$\int N^{-2/3} dN = 3N^{1/3} = 3N \times N^{-2/3} \quad .$$

Combining this with (S5) leads again to (S3).

7.4) CHEMICAL POTENTIAL VERSUS TEMPERATURE

The number of particles, given by

$$N = \int_0^{\infty} \mathcal{D}(\epsilon) f(\epsilon) d\epsilon \quad , \quad (S1)$$

must be independent of temperature. We consider two energy levels symmetrically displaced from $\epsilon = \mu$, $\epsilon_+ = \mu + \delta$ and $\epsilon_- = \mu - \delta$, and use the symmetry property (6.74) of the Fermi-Dirac distribution function:

$$f(\mu + \delta) = 1 - f(\mu - \delta) \quad . \quad (S2)$$

If the density of states $\mathcal{D}(\varepsilon)$ increases with increasing energy, as in 3 dimensions, this property implies that there are more occupied states at ε_+ than holes at ε_- . The opposite is true if $\mathcal{D}(\varepsilon)$ decreases with increasing energy as in 1 dimension. If the chemical potential μ did not change with temperature, the value of the integral in (S1) would increase with temperature if $\mathcal{D}(\varepsilon)$ increases with energy. To compensate for such an increase, μ must decrease. Similarly, if $\mathcal{D}(\varepsilon)$ decreases with energy, a temperature-independent μ would imply a decrease of N with increasing temperature. To compensate for such a decrease, μ must increase.

There are numerous different ways to display this behavior graphically.

Extensions. The rate of change of the chemical potential is easily determined quantitatively by writing (S1) in the form

$$N = \int_0^{\mu} \mathcal{D}(\varepsilon) d\varepsilon + \int_0^{\infty} [\mathcal{D}(\mu+\delta) - \mathcal{D}(\mu-\delta)] f(\mu+\delta) d\delta \quad , \quad (S3)$$

with the aid of (S2). If $\tau \ll \mu$, which is the case of greatest interest, we may approximate the square brackets as $2(d\mathcal{D}/d\varepsilon)\delta$, with the derivative taken at $\varepsilon = \mu$. The second integral in (S2) then becomes

$$2(d\mathcal{D}/d\varepsilon) \int_0^{\infty} f(\mu+\delta) \delta d\delta = 2\tau^2 (d\mathcal{D}/d\varepsilon) \int_0^{\infty} \frac{x dx}{e^x + 1} \quad ,$$

where we have inserted (6.4) for $f(\varepsilon)$. The integral has the value $\pi^2/12$ -- see (p. 192). Insertion of this result into (S3) and differentiation yields

$$0 = \mathcal{D}(\mu) d\mu/d\tau + (\pi^2 \tau/3) (d\mathcal{D}/d\varepsilon)$$

or

$$d\mu/d\tau = -(\pi^2\tau/3)(d \log \mathcal{D}/d\varepsilon) \quad .$$

In 3 dimensions, from (19), $\mathcal{D}(\varepsilon) \propto \varepsilon^{3/2}$. From this $d \log \mathcal{D}/d\varepsilon = 1/2\varepsilon > 0$, hence

$$d\mu/d\tau = -\pi^2\tau/6\varepsilon_F < 0 \quad , \quad (S4)$$

where the logarithmic derivative has been evaluated at $\varepsilon = \varepsilon_F$. In 1 dimension, from (86), $\mathcal{D}(\varepsilon) \propto \varepsilon^{-1/2}$, which leads to a reversal of the sign in (S4).

7.5) LIQUID ^3He AS A FERMI GAS

(a) With $M(^3\text{He}) \cong 3 \text{ amu} \cong 4.98 \times 10^{-24} \text{ g}$ we have $N/V = 0.081 \text{ g cm}^{-3}/3 \text{ amu} \cong 1.63 \times 10^{22} \text{ cm}^{-3}$. Next, from (40):

$$\varepsilon_F = (\hbar^2/2m)(3\pi^2 N/V)^{2/3} = 6.86 \times 10^{-16} \text{ erg} \quad ,$$

$$T_F = \varepsilon_F/k_B = 4.97 \text{ K} \quad ,$$

$$v_F = (2\varepsilon_F/M)^{1/2} = 1.66 \times 10^4 \text{ cm s}^{-1} \quad .$$

(b) From (38),

$$C_V = (\pi^2/2)N_A k_B T/T_F = 0.99 \text{ K} \quad ,$$

which is only $1/(2.91)$ of the observed value. The motion of individual ^3He atoms in liquid ^3He is correlated. To move a ^3He atom requires pushing other atoms out of the way. The kinetic energy of the other atoms getting out of the way is part of the kinetic energy involved in moving the atom, making it appear as if each ^3He atom had an effective mass greater than its real mass. The heat capacity suggests an effective mass 2.91 times $M(^3\text{He})$.

7.6) MASS-RADIUS RELATIONSHIP FOR WHITE DWARFS

(a) See solution to problem 4.3 for this part.

(b) From (10) and (7), with $n = N/V$:

$$U_0 = (3/5)N\varepsilon_F = (3\hbar^2/10m)(3\pi^2/V)^{2/3} N^{5/3}.$$

With $N = N_e = N_H = M/M_H$, $V = 4\pi R^3/3$:

$$\begin{aligned} U_0 &= (3/10)(9\pi/4)^{2/3}(\hbar^2/mR^2)(M/M_H)^{5/3} \\ &= 1.10(\hbar^2/mR^2)(M/M_H)^{5/3}. \end{aligned}$$

(c) We set $3GM^2/5R = U_0$:

$$\begin{aligned} 3GM^{6/3}/5R &= 1.10(\hbar^2/mR^2)(M/M_H)^{5/3}, \\ C \equiv RM^{1/3} &= 1.84\hbar^2/GmM_H^{5/3} = 1.45 \times 10^{20} g^{1/3} \text{ cm}. \end{aligned}$$

(d) $\rho = M/(4\pi R^3/3) = 3M^2/4\pi C^3$. For $M = 2 \times 10^{33} g$: $\rho = 3.15 \times 10^5 g \text{ cm}^{-3}$. The corresponding radius is $R = C/M^{1/3} \cong 1.15 \times 10^9 \text{ cm}$.

(e) In a neutron star, the kinetic energy in (b) involves the neutron mass ($\cong M_H$) in place of the electron mass m . If we replace, in (c), m by $M_H = 1837m$, we obtain

$$C_n = (RM^{1/3})_n = C/1837 \cong 7.8 \times 10^{16} g^{1/3} \text{ cm}.$$

For $M = 2 \times 10^{33} g$ we have $R = C_n/M^{1/3} \cong 6.25 \text{ km}$. The corresponding density is $\rho = 1.95 \times 10^{15} g/\text{cm}^3$.

7.7) PHOTON CONDENSATION

We solve (4.48) for $\tau^3 = (k_B T_C)^3$:

$$\tau^3 = (\hbar c)^3 \pi^2 (N/V)/2.404; T_C = (\hbar c/k_B) \times (\pi^2 N/2.404V)^{1/3}.$$

For $N/V = 10^{20} \text{ cm}^{-3}$ we have $T_C = 1.70 \times 10^6 \text{ K}$.

7.8) ENERGY, HEAT CAPACITY, AND ENTROPY OF DEGENERATE BOSON GAS

Only the uncondensed bosons contribute to the energy. Their number $N_e(\tau)$ was calculated in (68); the expression for their energy differs from (68) only by an additional factor $\varepsilon = \tau x$:

$$U(\tau) = \frac{V}{4\pi^2} \left(\frac{2M}{\hbar^2} \right)^{3/2} \tau^{5/2} \int_0^\infty \frac{x^{3/2} dx}{e^x - 1} = AN_e(\tau)\tau \quad , \quad (S1)$$

where A is the ratio of the integral in (S1) to the integral (69). Numerically, $A = 0.770$ (see below). We use (73) to write $U(\tau)$ in the final form

$$U(\tau) = AN\tau^{5/2}\tau_E^{-3/2} \quad .$$

From this:

$$C_V(\tau) = (\partial U / \partial \tau)_{V,N} = \frac{5}{2} AN(\tau/\tau_E)^{3/2} \quad .$$

Because of (3.17), $C_V = \tau(\partial \sigma / \partial \tau)$:

$$(\partial \sigma / \partial \tau)_{V,N} = \frac{5}{2} AN\tau^{1/2}\tau_E^{-3/2} \quad ,$$

$$\sigma(\tau) = \frac{5}{3} AN(\tau/\tau_E)^{3/2} \quad ,$$

where in the last line we have used $\sigma(0) = 0$.

Comment. The integral in (S1) may be evaluated by the method in the footnote on p.204 of the text:

$$\int_0^\infty \frac{x^{3/2} dx}{e^x - 1} = \left(\sum_{s=1}^\infty s^{-5/2} \right) \int_0^\infty y^{3/2} e^{-y} dy \quad .$$

The infinite sum is easily summed to 1.341 on a programmable calculator. The integral is a gamma function (see Appendix A): $\Gamma(5/2) = 3/2 \Gamma(3/2)$, where $\Gamma(3/2)$ is the integral in the footnote on p.204. Hence

$$A = (3/2) \times 1.341 / 2.612 = 0.7702 \quad .$$

7.9) BOSON GAS IN ONE DIMENSION

In one dimension, the density of orbitals $\mathcal{Q}(\varepsilon)$ in the integral in (66) is not given by (65) but by an expression

that differs from (86) only by different values for the particle mass and for the spin multiplicity. For the derivation of (86) see solution to Problem 7.1. We pick up the trail with the form (86), and we note that $\mathcal{D}_1(\varepsilon) \propto \varepsilon^{-1/2}$. If this is inserted into the integral in (66), for $\lambda \rightarrow 1$:

$$N_e \propto \int_0^{\infty} \frac{1}{\varepsilon^{1/2}} \frac{d\varepsilon}{\exp(\varepsilon/\tau) - 1}.$$

When $\varepsilon \ll \tau$ the exponential may be expanded: $\exp(\varepsilon/\tau) - 1 \cong \varepsilon/\tau$. Hence for small values of ε/τ the integrand becomes proportional to $\varepsilon^{-3/2}$, which leads to an integral that diverges at its lower limit.

7.10) RELATIVISTIC WHITE DWARF STARS

(a) We take over the result (88,89) for the energy of the extreme relativistic electron Fermi gas. If we write $n = N/V = 3N/4\pi R^3$, this result becomes

$$\begin{aligned} U_0 &= \frac{3}{4} N \varepsilon_F = \frac{3}{4} N \hbar c (9N/4\pi R^3)^{1/3} \\ &= \frac{3}{4} \left(\frac{9\pi}{4} \right)^{1/3} \frac{\hbar c N^{4/3}}{R}. \end{aligned}$$

We equate this to the magnitude of the gravitational potential energy, assuming a uniform density:

$$\langle \text{P.E.} \rangle = \frac{3}{5} \frac{GM^2}{R} = \frac{3}{5} \frac{GM_H^2}{R} \frac{N^{6/3}}{R}.$$

(See Problems 4.1 and 7.6). The radius cancels out and we obtain the relation for N

$$N^{2/3} = \frac{5}{4} \left(\frac{9\pi}{4} \right)^{1/3} \frac{\hbar c}{GM_H^2}.$$

(b) $N = 8.4 \times 10^{57}$. If the Sun were all hydrogen: $N = M/M_H = 2 \times 10^{33} \text{ g}/M_H = 1.3 \times 10^{57}$.

7.11) FLUCTUATIONS IN A FERMI GAS

By definition, $\langle (\Delta N)^2 \rangle = \langle (N - \langle N \rangle)^2 \rangle = \langle N^2 \rangle - \langle N \rangle^2$. For fermions, only $N = 0, 1$ are allowed. For those values, $N^2 = N$ and $\langle N^2 \rangle = \langle N \rangle$. Therefore:

$$\langle (\Delta N)^2 \rangle = \langle N \rangle - \langle N \rangle^2 = \langle N \rangle (1 - \langle N \rangle) .$$

This result may also be obtained less elegantly but perhaps more systematically from (5.83) -- see solution to Problem 5.20 -- by inserting (6.4) for $\langle N \rangle = f(\varepsilon)$.

7.12) FLUCTUATION IN A BOSE GAS

The trick $\langle N^2 \rangle = \langle N \rangle$ of the preceding problem is not applicable to bosons. We draw on (5.83) -- see solution to Problem 5.10 -- by inserting (6.10) for $\langle N \rangle = f(\varepsilon)$:

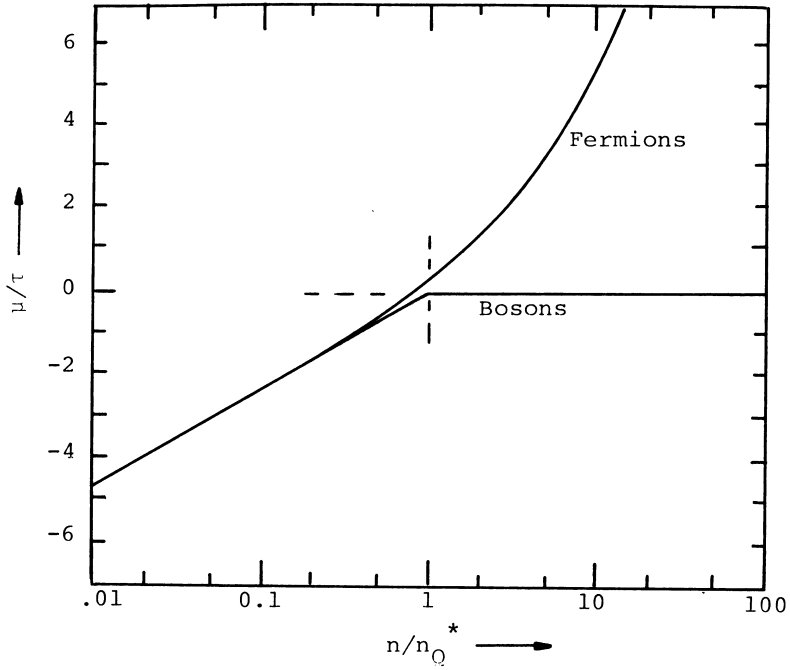
$$\begin{aligned} \langle (\Delta N)^2 \rangle &= \tau \partial \langle N \rangle / \partial \mu = \tau \frac{\partial}{\partial \mu} \frac{1}{\exp[(\varepsilon - \mu)/\tau] - 1} . \\ &= \frac{\exp[(\varepsilon - \mu)/\tau]}{\{\exp[(\varepsilon - \mu)/\tau] - 1\}^2} = \langle N \rangle (1 + \langle N \rangle) . \end{aligned}$$

7.13) CHEMICAL POTENTIAL VERSUS CONCENTRATION

(a) In the classical regime $n \ll n_Q$, and μ is given for both bosons and fermions by (6.48), which may be written as

$$\mu/\tau = \log(n/n_Q) - \log Z_{\text{int}} = \log(n/n_Q^*) . \quad (S1)$$

Here Z_{int} is the spin multiplicity (see (6.51)), and we have defined $n_Q^* = n_Q Z_{\text{int}}$, as in the solution to Problem 6.9. The form (S1) suggests that the most useful plot is a semilogarithmic plot (μ versus $\log n$), with μ and n expressed as multiples of τ and n_Q^* . (See drawing next page). In the quantum regime, bosons satisfy (55), $\mu/\tau = -1/N$. For any macroscopic volume in the quantum regime, $N = nV \gg n_Q V$ must be a large number, hence μ/τ is very close to zero, as shown.



(b) We restrict ourselves to spin- $\frac{1}{2}$ fermions. In the classical regime (S1) holds again, with the appropriate values of n_Q^* . With a spin multiplicity of 2, from (3.63):

$$n_Q^* = 2n_Q = 2(M\tau/2\pi\hbar^2)^{3/2}. \quad (S2)$$

In the quantum regime, from (7), with m replaced by M ,

$$\mu(\tau) \cong \mu(0) = \varepsilon_F = (\hbar^2/2M)(3\pi^2n)^{2/3}.$$

With (S2) this may be written

$$\mu/\tau = (9\pi/16)^{1/3} (n/n_Q^*)^{2/3} = 1.209(n/n_Q^*)^{2/3}.$$

An excellent approximation in the intermediate regime $n \cong n_Q$ is the Joyce-Dixon approximation discussed in Chapter 13. See (13.38) and Fig. 13.4, where $\eta \equiv \mu/\tau$, $r \equiv n/n_Q^*$. Our curve was taken from that figure.

7.14) TWO ORBITAL BOSON SYSTEM

From (53) and (54), or from (6.10):

$$\exp[(\varepsilon - \mu)/\tau] = 1 + 1/f(\varepsilon) \quad , \quad (S1)$$

$$\exp(-\mu/\tau) = 1 + 1/f(0) \quad . \quad (S2)$$

We set $f(0) = 2N/3 \gg 1$, $f(\varepsilon) = N/3 \gg 1$ and divide (S1) by (S2),

$$\exp(\varepsilon/\tau) = (1+3/N)/(1+3/2N) \cong 1+3/2N \quad ,$$

where in the last equality we utilized $3/2N \ll 1$. Taking logarithms, and again using $3/2N \ll 1$:

$$\varepsilon/\tau = \log(1+3/2N) \cong 3/2N; \quad \tau \cong 2N\varepsilon/3 \quad .$$

Note that $N\varepsilon$ occurs where we might naively have expected a quantity of order ε . This example illustrates once again a point made in the text, that the Bose-Einstein condensation of the majority of particles into the lowest orbital takes place at a temperature much larger than the energy separation between the lowest and next-lowest orbitals.